

Tables, Modules, Services & APIs

Extracted from the backend design · for the developers building it

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1. Database tables

PostgreSQL 16 with TimescaleDB. Every business table carries `tenant_id` and is protected by row-level security; primary keys are UUIDs, and the codes users see (MTR-1001, GW-01, AR-7F2 ...) are separate unique columns. All tables also have `created_at` / `updated_at`.

Area	Tables
Platform and tenants	<code>platform_admins</code> , <code>tenants</code> , <code>tenant_status_history</code> , <code>tenant_limits</code>
Identity and access	<code>users</code> , <code>user_sites</code> , <code>invites</code> , <code>auth_sessions</code> , <code>otp_codes</code> , <code>roles</code> , <code>permissions</code> , <code>role_permissions</code> , <code>field_permissions</code>
Sites, devices and meters	<code>sites</code> , <code>devices</code> , <code>device_credentials</code> , <code>register_maps</code> , <code>cost_centres</code> , <code>meters</code> , <code>energy_targets</code> , <code>virtual_meter_members</code> , <code>connection_flows</code>
Telemetry (time-series)	<code>readings</code>
Tariffs, shifts and schedules	<code>tariffs</code> , <code>tariff_instances</code> , <code>shifts</code> , <code>shift_schedules</code> , <code>shift_assignments</code> , <code>shift_instances</code>
Alerts and notifications	<code>alert_rules</code> , <code>alert_rule_state</code> , <code>alerts</code> , <code>alert_events</code> , <code>notifications</code> , <code>notification_preferences</code>
Audit, exports and device events	<code>audit_log</code> , <code>export_jobs</code> , <code>device_messages</code> , <code>device_commands</code> , <code>publish_destinations</code> , <code>publish_attempts</code> , <code>device_events</code>

1.1 Platform and tenants

platform_admins Control Center accounts — a separate realm from tenant users

Column	Type	Notes
<code>id</code>	uuid PK	
<code>username</code>	citext unique	e.g. superadmin
<code>name</code> , <code>email</code>	text	
<code>password_hash</code>	text	argon2id
<code>mfa_secret</code>	text null	TOTP; required in production
<code>last_login_at</code>	timestamptz null	

tenants One organisation (the Org ID users type at sign-in)

Column	Type	Notes
<code>id</code>	varchar(12) PK	Org ID: chosen slug (FEBINO) or generated 8-char code (HX4R7TQM)
<code>name</code>	text	legal name, Febino Solutions Pvt Ltd
<code>slug</code>	citext unique	Workspace URL, febino; <code>^[a-z0-9][a-z0-9-]{1,29}\$</code>
<code>industry</code>	text	Manufacturing plant, Data centre, Solar / renewable plant ...
<code>gst_in</code>	varchar(15) null	hidden from some roles by field permissions
<code>country</code> , <code>timezone</code> , <code>address</code> , <code>city</code>	text	timezone default Asia/Kolkata
<code>discom</code>	text	utility, e.g. BESCO, TANGEDCO
<code>contract_demand_kva</code>	integer	for maximum-demand checks
<code>meter_range</code>	text	1-10 11-50 51-200 200+ (sets default meter limit)
<code>plan</code>	enum	starter growth enterprise
<code>billing_cycle</code>	enum	monthly yearly
<code>contact_email</code> , <code>phone</code>	text	
<code>status</code>	enum	pending active suspended
<code>status_reason</code>	text null	e.g. Subscription invoice overdue
<code>suspended_until</code>	timestamptz null	suspension lifts itself at this time (scheduler)
<code>approved_at</code> , <code>approved_by</code>	timestamptz, uuid null	platform admin who approved
<code>currency</code>	char(3)	INR USD AED
<code>energy_unit</code>	enum	kWh MWh (display only; storage is always kWh)
<code>co2_factor</code>	numeric(5,3)	kg CO2 per kWh, default 0.820

Column	Type	Notes
default_tariff	numeric(8,2)	₹/kWh used when no ToD slot matches (UI demo uses 8)
deleted_at	timestampz null	soft delete; data purged after 30 days

tenant_status_history Every approve / reject / suspend / reactivate

Column	Type	Notes
id	uuid PK	
tenant_id	FK tenants	
from_status, to_status	enum	
reason	text null	
until	timestampz null	for temporary suspension
by_admin_id	FK platform_admins null	null = system (auto-lift)
at	timestampz	

tenant_limits Overrides of the plan's default limits (Control Center → Limits)

Column	Type	Notes
tenant_id	FK, PK part	
key	enum, PK part	users meters subMeters devices sites tariffs shifts schedules alertRules roles
value	integer	0 = unlimited
reason	text null	why the admin changed it
set_by, set_at	uuid, timestampz	

1.2 Identity and access

users A person's account inside one tenant (the same email may exist in several tenants)

Column	Type	Notes
id	uuid PK	
tenant_id	FK tenants	
emp_id	varchar(16)	EMP-014; unique per tenant, auto-next
first_name, last_name	text	
email	citext	unique per tenant
username	citext unique	unique across the platform (sign-in by username works without guessing tenant)
mobile	varchar(20) null	for OTP sign-in and SMS alerts
role_id	FK roles	
shift_id	FK shifts null	Shift A / B / C / General
department, designation	text null	
status	enum	invited active inactive suspended
two_factor	boolean	require OTP after password
language	varchar(8)	en-IN hi ta (My profile)
timezone	text null	IANA name; null = tenant's
password_hash	text null	null until the invite is accepted
last_active_at	timestampz null	shown as "12 min ago"
invited_by	FK users null	

user_sites Optional: restrict a user to some sites

Column	Type	Notes
user_id	FK, PK part	
site_id	FK, PK part	no rows = all sites

invites Invitation links (accept-invite.html?token=...)

Column	Type	Notes
id	uuid PK	
user_id	FK users	
token_hash	char(64) unique	SHA-256 of the random token; the token itself is only in the email
invited_by	FK users	
expires_at	timestamp tz	7 days
accepted_at	timestamp tz null	

auth_sessions Signed-in devices (Settings → Your active sessions)

Column	Type	Notes
id	uuid PK	
realm	enum	tenant platform
user_id / admin_id	uuid	one of them
tenant_id	FK null	current tenant (changes on switch-tenant)
refresh_token_hash	char(64)	rotated on every refresh; reuse of an old one revokes the session
remember	boolean	true: 30 days, false: 12 hours (platform: 8 hours)
ip, user_agent, device_label	text	"Chrome on Windows · Bengaluru"
last_used_at, expires_at, revoked_at	timestamp tz	

otp_codes One-time codes (verify email, OTP sign-in, password reset, login approval)

Column	Type	Notes
id	uuid PK	
purpose	enum	verify_email login_otp reset_password login_approval
target	text	email or mobile
user_id / tenant_id	uuid null	
code_hash	text	6 digits, hashed
expires_at	timestamp tz	10 minutes
attempts	smallint	locked after 5 wrong tries
consumed_at	timestamp tz null	

roles Built-in and custom roles

Column	Type	Notes
id	uuid PK	
tenant_id	FK tenants	
name	text	Tenant Admin, Energy Manager, Supervisor, Operator, Technician, Viewer, Auditor, or custom
description	text	
is_system	boolean	Type: Built-in Custom; built-in roles cannot be deleted; Tenant Admin cannot be restricted
landing_page	text	Default landing page: dashboard meters analytics alerts

permissions Catalogue of permission keys (Roles → Permissions tab)

Column	Type	Notes
key	varchar(48) PK	module.action, e.g. meters.export, alerts.acknowledge
module	enum	dashboard, meters, devices, connections, analytics, alerts, shifts, users, roles, settings
action	text	view, create, edit, delete, export, approve, acknowledge ...
description	text	What does this allow?
tenant_id	FK null	null = platform-defined; tenants may add custom keys

role_permissions Which keys a role has (the Permission matrix is a view over this)

Column	Type	Notes
role_id	FK, PK part	
permission_key	FK permissions, PK part	

field_permissions Field-level visibility per role (field-perms.js)

Column	Type	Notes
tenant_id, role_id	FK, PK part	
module, field	text, PK part	e.g. meters.cost, devices.mac, users.mobile, settings.gstin
can_view, can_edit	boolean	edit implies view; no row = full access

1.3 Sites, devices and meters

sites A location with its own utility connection

Column	Type	Notes
id	uuid PK	
tenant_id	FK	
name	text	Plant 1 · Bengaluru
address	text	
utility, consumer_no	text	DISCOM and service number
contract_demand_kva	integer	
default_tariff	enum	tod flat
flat_rate	numeric(8,2) null	₹/kWh when default_tariff = flat
timezone	text null	falls back to the tenant's
status	enum	active inactive

devices Gateways and data loggers (Device Management)

Column	Type	Notes
id	uuid PK	
tenant_id, site_id	FK	
code	varchar(16)	GW-01; unique per tenant
name	text	Main LT Panel Gateway
type	enum	modbus_tcp_gateway data_logger edge_4g lorawan_gateway
model, serial	text	serial hidden from Operator/Viewer by default
location	text	
ip, port, mac	inet, int, macaddr	
link	enum	ethernet wifi lte
field_protocol	text	Modbus RTU (RS-485), Modbus TCP, LoRaWAN
uplink	enum	mqtt_push https_push modbus_tcp_poll
polling_s	integer	15, 30, 60

Column	Type	Notes
baud	integer null	9600 19200 38400; only for RS-485
time_sync	enum	ntp server
firmware, firmware_available	text	an update is offered when they differ
installed_on	date	
status	enum	online weak offline maintenance (derived, see 6.6)
signal_pct	smallint null	
last_seen_at	timestamptz null	"4 s ago"

device_credentials The device token — how a device proves who it is

Column	Type	Notes
device_id	FK PK	
client_id	text unique	{orgId}.{deviceCode}, e.g. FEBINO.GW-07 (MQTT username)
token_hash	char(64) unique	SHA-256 of the token; the token itself is never stored
token_prefix	char(8)	"edt_" + first 4 chars — to recognise a token in logs
token_last4	char(4)	shown masked: edt_....a91F
created_at, created_by	timestamptz, FK users	generated together with the device
rotated_at, rotated_by	timestamptz null	Edit device → Regenerate token
last_used_at	timestamptz null	
cert_fingerprint	text null	optional mutual TLS instead of a token

register_maps Modbus register layout per meter model (for polled devices)

Column	Type	Notes
id	uuid PK	
model	text unique	Schneider PM5560, Secure Elite 440, L&T WL4405 ...
registers	jsonb	[{param:'kw', address:3060, type:'float32', order:'ABCD', scale:1}, ...]
is_global	boolean	shared library; tenants may add their own

cost_centres For allocating energy cost (Analytics, sub-meters)

Column	Type	Notes
id	uuid PK	
tenant_id	FK	
code	varchar(16)	CC-PROD, CC-UTIL, CC-ADMIN, CC-IT, CC-FAC; unique per tenant
name	text	Production, Utilities ...

meters Energy meters and sub-meters (one table, parent_id makes a sub-meter)

Column	Type	Notes
id	uuid PK	
tenant_id, site_id	FK	
code	varchar(20)	MTR-1001 for meters, SM-1001-01 for sub-meters; unique per tenant
parent_id	FK meters null	null = main meter (counts toward 'meters' limit); set = sub-meter ('subMeters' limit)
name	text	Main Incomer, Feeder A · Production Line 1
kind	enum	consumer generator (solar, DG) — generators show 'Generating'
category	enum	incomer hvac lighting process utility other
cost_centre_id	FK cost_centres null	mainly for sub-meters
model	text	links to register_maps.model
location	text	
phase	smallint	1 = single-phase 2-wire 230 V L-N; 3 = three-phase 4-wire 415 V L-L

Column	Type	Notes
nominal_v	numeric	230 or 415 (for formulas and limits)
ct_primary, ct_secondary	integer null	CT 400/5 → 400, 5; null = direct (e.g. Direct 63 A)
direct_a	integer null	rating for direct-connected sub-meters
rated_kw	numeric null	for Load % = kW / rated
device_id	FK devices null	the device that reads it (may differ from the parent's device)
modbus_slave_id	smallint null	1–247; unique together with device_id
register_map_id	FK register_maps null	defaults from model
register_offset	integer null	multi-channel meters: channel's base register (sub-meter form)
mode	enum	active maintenance disabled — the only status a user sets
icon, color	text	UI only (i-cpu, --cyan)
status	enum	online idle offline maintenance generating standby — derived from readings + mode, never edited
archived_at	timestamptz null	keeps history when a meter is removed

energy_targets Monthly target & allocation (Analytics → Edit target & allocation)

Column	Type	Notes
id	uuid PK	
tenant_id, meter_id	FK	
month	date	first day of month; unique (meter_id, month)
target_kwh	numeric(12,1)	> 0
cost_centre_id	FK null	
allocation_pct	numeric(5,2)	0–100; share charged to the cost centre
tariff_mode	enum	tod flat
notes	text null	reason for target change
set_by	FK users	

virtual_meter_members Virtual meters (Σ nodes on the Connections canvas)

Column	Type	Notes
virtual_id	FK meters (kind=virtual)	a virtual meter is a meters row without device
member_id	FK meters	a meter, sub-meter or another virtual meter
sign	smallint	+1 add, -1 subtract (e.g. Incomer – Solar)

connection_flows Saved and deployed versions of the Connections canvas

Column	Type	Notes
id	uuid PK	
tenant_id	FK	
version	integer	increments on each deploy
nodes	jsonb	[[id, type: submeter meter virtual device, name, ref, x, y, props]]
wires	jsonb	[[from, to, kind: direct sub link]]
is_draft	boolean	one draft per tenant; deploy creates a new version
deployed_by, deployed_at	uuid, timestamptz	

1.4 Telemetry (time-series)

readings TimescaleDB hypertable — one row per meter per reading

Column	Type	Notes
time	timestamptz	meter/gateway timestamp (UTC); chunk interval 1 day
tenant_id	varchar(12)	for RLS and per-tenant queries
meter_id	uuid	PK = (meter_id, time) — duplicates are ignored
kw, kvar, kva	real	total active / reactive / apparent power
pf	real	total power factor (−1...1)
v1, v2, v3	real	phase-to-neutral volts; 1φ meters fill only v1
v12, v23, v31	real null	line-to-line volts (3φ only)
i1, i2, i3, i_n	real	phase and neutral amps; 1φ meters fill only i1
hz	real	frequency
thd_v, thd_i	real	% total harmonic distortion
kwh_imp, kwh_exp	double precision	cumulative energy registers (import / export)
kvarh	double precision null	
quality	smallint	bit flags: 1 estimated, 2 out-of-range, 4 register reset, 8 late
received_at	timestamptz	server time (for ingest-lag monitoring)

1.5 Tariffs, shifts and schedules

tariffs Time-of-Day tariff slots T1–T5 (Settings → Tariff slabs)

Column	Type	Notes
id	uuid PK	
tenant_id, site_id	FK (site null = all sites)	
slot	varchar(4)	T1 ... T5
name	text	Off-peak (night), Morning peak, Evening peak ...
start_time, end_time	time	22:00 → 06:00 crosses midnight
days	smallint	bit mask Mon=1 ... Sun=64 (Mon – Sat = 63)
energy_rate	numeric(8,2)	₹ per kWh, e.g. 6.80, 9.20, 9.60
demand_charge	numeric(8,2)	₹ per kVA of maximum demand per month
valid_from, valid_to	date	keeps expired slabs (FY 2025-26) for history
status	derived	current if today is within valid_from...valid_to, otherwise expired (not stored)

tariff_instances Each run of a ToD slot (Energy cost → Instances)

Column	Type	Notes
id	uuid PK	
tenant_id, site_id	FK	
code	varchar(20)	TI-1008-T2; TI-1005-T1E = Monday's early T1 after Sunday flat
tariff_id	FK tariffs	the slab in force when it ran (rate is copied below)
starts_at, ends_at	timestamptz	T1 22:00 → 06:00 is one run, counted on the day it starts
status	enum	upcoming running closed
rate	numeric(8,2)	₹/kWh snapshot
energy_kwh, cost, peak_kw	numeric null	updated live while running, final when closed

shifts Shift definitions (Shift page)

Column	Type	Notes
id	uuid PK	
tenant_id	FK	
code, name	text	A / Shift A

Column	Type	Notes
start_time, end_time	time	end ≤ start means it ends the next day
break_min	smallint	0 30 45 60
count_on	enum	start_day end_day — which day an overnight shift belongs to
supervisor_id	FK users null	
grace_min	smallint	0–60 late-arrival grace
energy_target_kwh	numeric null	per shift run
status	enum	active inactive
color	text	amber orange violet cyan green

shift_schedules Repeating plans (Schedule page)

Column	Type	Notes
id	uuid PK	
tenant_id	FK	
name	text	
pattern	jsonb	weekly or rotation: [{day/offset, shift_id}]
shift_id	FK	the form's Shift
days	smallint	Repeats on (bit mask) — the days a shift runs come from its schedules
pattern	jsonb null	rotations later: [{offset, shift_id}]
starts_on, ends_on	date	Effective from / Until
status	enum	active paused

shift_assignments Who works which shift (Schedule → Assign staff to shift)

Column	Type	Notes
id	uuid PK	
tenant_id, user_id, shift_id	FK	
schedule_id	FK null	
from_date, to_date	date	to ≥ from; one user cannot have overlapping assignments

shift_instances Each actual run of a shift (Instance page)

Column	Type	Notes
id	uuid PK	
tenant_id, shift_id, schedule_id	FK	
code	varchar(20)	SI-1008-A
starts_at, ends_at	timestamptz	planned times, overnight resolved
actual_start, actual_end	timestamptz null	Planned / actual column
staff_count	smallint	from shift_assignments
production_units	numeric null	entered by supervisor; kWh / unit = energy_kwh / production_units
status	enum	planned running closed
energy_kwh, cost, peak_kw, avg_pf	numeric null	snapshot written when the instance closes

1.6 Alerts and notifications

alert_rules Per-meter rules (meter page → Alert rules; same shape the UI already saves)

Column	Type	Notes
id	uuid PK	
tenant_id	FK	
meter_id	FK null	null for tenant-wide rules (Alerts page)

Column	Type	Notes
code	varchar(10)	AR-7F2
name	text	1–80 chars
severity	enum	critical warning info
enabled	boolean	
scope	enum	meter subs meter_and_sub all all_meters all_sub
sub_ids	uuid[]	selected sub-meters ([] = all)
evaluate	enum	live 15m 1h shift day — live checks every reading; others check the window when it closes
cond	jsonb	{mode: guided expression, match: all any, rows:[{param, op, v1, v2}], expr, hold}
active	jsonb	{mode: always shifts slots custom, shifts, slots, from, to, days}
notify	jsonb	{channels: app email sms whatsapp webhook, roles, users, emails, repeat, resolve, escalate}
compiled	jsonb	server-validated form of cond (parsed expression, variable list)
created_by	FK users	

alert_rule_state Where each rule stands for each meter it watches (also cached in Redis)

Column	Type	Notes
rule_id, meter_id	PK	
state	enum	ok pending firing
since	timestampz	when the current state began
last_value	jsonb	values that made it true
alert_id	FK alerts null	the open alert while firing
last_notified_at, escalated_at	timestampz null	

alerts The Alert log

Column	Type	Notes
id	uuid PK	
tenant_id	FK	
rule_id	FK null	null for system alerts (device offline, firmware, budget)
meter_id, device_id	FK null	
title	text	THD above 8%
severity	enum	
category	enum	threshold power_quality energy device communication info
value_text, limit_text	text	9.1% / 8%
status	enum	open acknowledged resolved
assigned_to	FK users null	Update alert → Assign to
raised_at, acknowledged_at, resolved_at	timestampz	
acknowledged_by, resolved_by	FK users null	null = auto-resolved
note	text null	Resolution note

alert_events Timeline of an alert (raised, notified, escalated, acknowledged, resolved)

Column	Type	Notes
id	bigserial	
alert_id	FK	
type	text	
detail	jsonb	

Column	Type	Notes
at	timestampz	

notifications Every message sent, and the bell-icon inbox

Column	Type	Notes
id	uuid PK	
tenant_id	FK	
user_id	FK null	null for plain email addresses / webhooks
alert_id	FK null	
channel	enum	app email sms whatsapp webhook
to	text	
title, body	text	
status	enum	queued sent failed
error	text null	
sent_at, read_at	timestampz null	read_at for in-app

notification_preferences Per user

Column	Type	Notes
user_id	FK PK	
channels	jsonb	per severity, e.g. critical: [app, sms]
quiet_hours	jsonb null	except critical

1.7 Audit, exports and device events

audit_log Who changed what (append-only)

Column	Type	Notes
id	bigserial	
tenant_id	varchar null	null for platform actions
actor_type, actor_id	enum, uuid	user platform_admin device system
action	text	user.invite, meter.update, flow.deploy, tenant.suspend ...
entity, entity_id	text	
before, after	jsonb	changed fields only; secrets removed
ip	inet	
at	timestampz	

export_jobs Large CSV / PDF / Excel exports run in the background

Column	Type	Notes
id	uuid PK	
tenant_id, user_id	FK	
kind	enum	energy alerts readings users shift_energy
format	enum	csv pdf xlsx
params	jsonb	{from, to, meters, columns}
status	enum	queued running done failed
file_key	text null	object-storage key
expires_at	timestampz	files deleted after 7 days

device_messages Telemetry log — every message per meter and polling cycle (hypertable, kept 7 days)

Column	Type	Notes
time	timestampz	read time (device clock)
tenant_id, device_id	FK	
meter_id	FK null	null when the slave is unknown
slave	smallint	Modbus address
status	enum	ok late timeout crc out_of_range
latency_ms	integer null	received_at – time; null for timeouts
payload	jsonb	the reading exactly as sent (Raw button)
received_at	timestampz	

device_commands Commands sent to a device (Commands tab)

Column	Type	Notes
id	uuid PK	
code	varchar(12)	CMD-3FT7
tenant_id, device_id	FK	
command	enum	ping get_info sync_time set_polling read_register clear_buffer restart
params	jsonb	{slave, fc, address, count} / {interval}
sent_by	FK users	needs devices.command permission
status	enum	queued sent success failed timeout (10 s without a reply)
reply	jsonb null	the device's answer
reply_ms	integer null	
created_at, sent_at, replied_at	timestampz	

publish_destinations Where a device's readings are forwarded (Data publish tab)

Column	Type	Notes
id	uuid PK	
tenant_id, device_id	FK	
name	text	unique per device
type	enum	mqtt https
url	text	mqtt://host:8883 or https://...
topic	text null	MQTT topic template with {org}, {device}, {meter}; no + or #
qos, retain	smallint, boolean	MQTT only
method	enum null	POST PUT (HTTPS)
auth_secret	text null	encrypted; only a masked hint (Bearer •••• 7f2a) is ever returned
interval	text	reading 60 300 900 seconds
format	enum	flat (one row per meter) grouped (one message per device)
fields	text[]	kw, kva, pf, v, i, hz, thd_v, thd_i, kwh
meter_ids	uuid[]	empty = all meters of the device
enabled	boolean	

publish_attempts Every publish (Data logs tab, together with device_commands) — hypertable, kept 30 days

Column	Type	Notes
time	timestampz	
destination_id	FK	
is_test	boolean	Test publish button
status	enum	success retried (delivered after a retry) failed (gave up after 3 tries)
detail	text	HTTP 200 / PUBACK · QoS 1 / HTTP 503 ...

Column	Type	Notes
duration_ms, bytes	integer	
payload	jsonb	the message sent or attempted

device_events Online/offline changes, firmware updates, reboots

Column	Type	Notes
id	bigserial	
device_id	FK	
type	text	online offline weak_signal firmware_update config
detail	jsonb	
at	timestampz	

2. Modules

A module is one area of the application. Each row shows the pages it covers, the backend services behind it, the tables it uses and the permission key (module in the role permission matrix) that controls access.

Module	Pages	Services	Tables	Permission
Authentication & onboarding	login, register, verify-email, register-success, forgot-password, accept-invite, login-approval, awaiting-approval	AuthService, TenantService (register), InviteService	users, auth_sessions, otp_codes, invites, tenants	— (public)
Control Center (platform)	admin-login, tenants, tenant, tenant-approvals	PlatformService, AuditService	platform_admins, tenants, tenant_status_history, tenant_limits, audit_log	platform realm
Dashboard	dashboard	AnalyticsService, LiveService + RealtimeGateway	readings_15m / 1h / 1d, meters, tariffs	dashboard
Energy meters	meters, meter (tabs: Live, Energy, Sub-meters, Alerts, Device & connections)	MeterService, TelemetryService, LiveService, AlertRuleService	meters, readings (+ aggregates), cost_centres, energy_targets, alert_rules	meters
Device management	devices, device (tabs: Overview, Meters, Telemetry, Commands, Data publish, Data logs, Connections)	DeviceService, IngestionService, TelemetryLogService, DeviceCommandService, PublishService	devices, device_credentials, register_maps, device_messages, device_commands, publish_destinations, publish_attempts, device_events	devices
Connections	connections	ConnectionService	connection_flows, meters, virtual_meter_members	connections
Analytics	analytics	AnalyticsService, TariffService + CostService, ExportService	readings aggregates, energy_targets, cost_centres, export_jobs	analytics
Alerts	alerts (and each meter's Alerts tab)	AlertRuleService, AlertEngine, AlertService, NotificationService	alert_rules, alert_rule_state, alerts, alert_events, notifications, notification_preferences	alerts
Shift management	shifts, shift-schedule, shift-instances	ShiftService	shifts, shift_schedules, shift_assignments, shift_instances	shifts
Energy cost	tariffs, tariff-instances	TariffService + CostService, TariffInstanceService	tariffs, tariff_instances	settings (tariffs)
Users	users	UserService + InviteService	users, user_sites, invites	users
Roles & permissions	roles	RoleService + PermissionService	roles, permissions, role_permissions, field_permissions	roles
Settings	settings (Organisation, Sites, Notifications, Security, Plan & billing, Appearance)	TenantService, SiteService, AuthService (sessions), NotificationService (preferences)	tenants, sites, auth_sessions, notification_preferences	settings
Cross-cutting	every page	AuditService, SchedulerService, ExportService, LiveService (WebSocket)	audit_log, export_jobs	—

3. Services

Each service is one backend module (NestJS) with a controller, business logic and a repository. Services marked worker run as separate processes so heavy traffic never slows page loads.

Service	Purpose	Main methods
AuthService	Sign-in, sessions and one-time codes	login, refresh, logout, me, requestOtp, verifyOtp, forgotPassword, resetPassword, switchTenant, listSessions, revokeSession
PlatformService (Control Center)	Tenant approvals, suspension and limits	listTenants, getTenant, approve, reject, suspend, reactivate, setLimits, deleteTenant
TenantService	Organisation profile, registration and limits	register, getProfile, updateProfile, getLimits, usage, check
UserService + InviteService	Team members	list, get, invite, update, remove, resendInvite, acceptInvite, nextEmpId
RoleService + PermissionService	Roles, module permissions and field permissions	listRoles, createRole, updateRole, deleteRole, getMatrix, setMatrix, getFieldPerms, setFieldPerms, can, canField
SiteService	Locations	list, create, update, remove
DeviceService	Gateways and loggers	list, get, create (→ device + token), update, remove, meters, regenerateToken, requestFirmwareUpdate
MeterService	Meters, sub-meters and virtual meters	tree, get, create, createSub, update, setMaintenance, archive
ConnectionService	Connections canvas	getFlow, saveDraft, validate, deploy, versions, rollback
IngestionService (worker)	Getting readings in	handleMqttMessage, handleHttpBatch, pollDevice, normalise, publish
TelemetryService	Storing and reading time-series	writeBatch, series, latest, energy, demand
LiveService + RealtimeGateway	Realtime to the browser	onReading, snapshot, emit
AnalyticsService	Reports and the Dashboard	dashboardSummary, consumption, subMeterReport, demand, powerQuality
TariffService + CostService	Money	list, create, update, remove, rateAt, cost
ShiftService	Shifts, schedules, instances	shifts CRUD, schedules CRUD, generateInstances, closeInstance, instanceEnergy
AlertRuleService	Rule management	list, create, update, remove, compile, test
AlertEngine (worker)	Evaluating rules 24×7	onReading, onWindowClose, evaluate, transition
AlertService	The Alert log	list, acknowledge, acknowledgeAll, resolve, remove
NotificationService (worker)	Delivery	enqueue, send, markRead, inbox
ExportService (worker)	CSV / PDF / Excel	create, status, download
SchedulerService	Timed jobs (BullMQ repeatable)	register jobs
AuditService	Traceability	record, list
DeviceCommandService	Commands to a device	send, list, get, onReply, timeout
PublishService (worker)	Forwarding readings to other systems	createDestination, updateDestination, removeDestination, test, preview, runInterval
TelemetryLogService	Device telemetry log	record, list, chartSeries, dataLogs
TariffInstanceService	Runs of Time-of-Day slots	generate, onReading, close, list, summary, get

AuthService — Sign-in, sessions and one-time codes

Responsibilities	• Sign in with Org ID + username / email / mobile + password; checks tenant status (pending → awaiting approval page, suspended → reason and date) and user status. • Optional 2FA: password then 6-digit OTP; OTP-only sign-in by mobile/email. • Issues a 15-minute access JWT and a rotating refresh token in an httpOnly cookie; remember-me 30 days, otherwise 12 hours. • Forgot / reset password, verify email, login approval, switch tenant (same email in another active tenant), list and revoke sessions.
Main methods	login, refresh, logout, me, requestOtp, verifyOtp, forgotPassword, resetPassword, switchTenant, listSessions, revokeSession

PlatformService (Control Center) — Tenant approvals, suspension and limits

Responsibilities	• Separate realm and 8-hour session for platform admins, MFA required. • Approve or reject pending registrations (emails the tenant admin), suspend with reason and optional end date, reactivate, delete. • Revokes every session of a suspended tenant immediately; scheduler lifts timed suspensions. • Edit limits per tenant with a reason; view a tenant's users and usage.
Main methods	listTenants, getTenant, approve, reject, suspend, reactivate, setLimits, deleteTenant

TenantService — Organisation profile, registration and limits

Responsibilities	• Self-registration creates a pending tenant + its admin user (generates an 8-character Org ID or reserves the chosen slug). • Profile settings (name, GSTIN, address, utility, contract demand, time zone, CO2 factor). • LimitService.check(tenant, key) is called inside the same transaction as every create — the server, not the browser, enforces limits (409 LIMIT_REACHED).
Main methods	register, getProfile, updateProfile, getLimits, usage, check

UserService + InviteService — Team members

Responsibilities	• List with search, role, status, department filters and server-side paging. • Invite (creates user in 'invited' state, emails a link), resend, accept (sets password), suspend, deactivate, delete. • Next employee ID, email unique per tenant, username unique platform-wide, site restrictions.
Main methods	list, get, invite, update, remove, resendInvite, acceptInvite, nextEmpId

RoleService + PermissionService — Roles, module permissions and field permissions

Responsibilities	• CRUD custom roles; built-in roles are seeded per tenant and cannot be deleted. • Permission matrix per module/action; field permissions per role (view/edit). • A guard checks module permissions on every route; a response interceptor removes fields the role may not view and rejects writes to fields it may not edit.
Main methods	listRoles, createRole, updateRole, deleteRole, getMatrix, setMatrix, getFieldPerms, setFieldPerms, can, canField

SiteService — Locations

Responsibilities	• CRUD sites with utility, consumer number and contract demand; limit 'sites'.
Main methods	list, create, update, remove

DeviceService — Gateways and loggers

Responsibilities	• CRUD devices; generates the device token on create and returns it once; regenerate token (old one revoked, device disconnected). • Lists the meters a device reads (with slave IDs); firmware info and update requests; status history. • Hidden fields (IP, MAC, serial) filtered by field permissions.
Main methods	list, get, create (→ device + token), update, remove, meters, regenerateToken, requestFirmwareUpdate

MeterService — Meters, sub-meters and virtual meters

Responsibilities	• Meter tree (main meters with their sub-meters), filters by site, status, consumer/generator. • Create meter / sub-meter with Supply (1φ / 3φ), CT ratio, rated kW, device and slave ID; limits 'meters' and 'subMeters'. • Validates that (device, slave ID) is free and that 1φ meters have no v2/v3 parameters in their rules. • Maintenance mode on/off; archive instead of hard delete.
Main methods	tree, get, create, createSub, update, setMaintenance, archive

ConnectionService — Connections canvas

Responsibilities	• Load registry-based default flow, save draft, validate, deploy (versioned), list versions, roll back.
Main methods	getFlow, saveDraft, validate, deploy, versions, rollback

IngestionService (worker) — Getting readings in

Responsibilities	• MQTT consumer, HTTPS ingest endpoint and Modbus TCP poller all produce the same normalised reading. • Authenticates the device, maps slave ID → meter via a cached device map, applies scaling, rejects impossible values, flags register resets. • Deduplicates on (meter, time), stamps received_at, appends to Redis Stream 'readings', updates device last_seen.
Main methods	handleMqttMessage, handleHttpBatch, pollDevice, normalise, publish

TelemetryService — Storing and reading time-series

Responsibilities	• DB writer consumer: batches stream entries and inserts with COPY (every 1 s or 500 rows). • Query API for charts: picks raw or the right aggregate, fills gaps, returns only requested parameters. • Daily jobs: shift instance snapshots, daily energy totals.
Main methods	writeBatch, series, latest, energy, demand

LiveService + RealtimeGateway — Realtime to the browser

Responsibilities	• Consumes the stream, updates live:{meter} in Redis, computes load %, imbalance, Σ sub-meters and Δ unmetered, today's kWh and cost. • Emits to Socket.IO rooms, at most once per second per meter, filtered by field permissions (e.g. no cost for Operators).
Main methods	onReading, snapshot, emit

AnalyticsService — Reports and the Dashboard

Responsibilities	• Dashboard summary (live load, today's kWh, cost, CO2, max demand vs contract, peak time). • Consumption by meter / sub-meter / site / shift / day, sub-meter report, power quality, solar vs forecast.
Main methods	dashboardSummary, consumption, subMeterReport, demand, powerQuality

TariffService + CostService — Money

Responsibilities	• CRUD ToD slots; splits energy by slot (in the site's time zone) and applies energy rate and demand charge.
Main methods	list, create, update, remove, rateAt, cost

ShiftService — Shifts, schedules, instances

Responsibilities	• CRUD shifts and schedules; generates shift instances 14 days ahead on the days each schedule repeats; resolves overnight shifts and the 'count on' day; closes instances and stores their energy.
Main methods	shifts CRUD, schedules CRUD, generateInstances, closeInstance, instanceEnergy

AlertRuleService — Rule management

Responsibilities	• CRUD rules with the UI's JSON shape; validates parameters against the meter's phase; compiles expressions safely; limit 'alertRules'. • Dry-run: replays the last 24 hours and reports when the rule would have fired.
Main methods	list, create, update, remove, compile, test

AlertEngine (worker) — Evaluating rules 24×7

Responsibilities	• Live rules on every reading, window rules when a window closes, 'no readings' rules from the offline checker. • State machine with hold time, active windows, repeat, auto-resolve and escalation (section 7). Writes alerts and alert_events, queues notifications.
Main methods	onReading, onWindowClose, evaluate, transition

AlertService — The Alert log

Responsibilities	• List/filter/page alerts, acknowledge, acknowledge all, resolve with note, delete (admin).
Main methods	list, acknowledge, acknowledgeAll, resolve, remove

NotificationService (worker) — Delivery

Responsibilities	• Resolves roles → users, applies preferences and quiet hours, sends via app (WebSocket + inbox), email, SMS, WhatsApp or signed webhook; retries with back-off; records status.
Main methods	enqueue, send, markRead, inbox

ExportService (worker) — CSV / PDF / Excel

Responsibilities	• Small exports are generated on request; large ranges (over 31 days or 50k rows) become a background job with a download link. Columns match the current exports (Date, Meter ID, Meter, Parent meter, Location, Phase, Energy, Peak, Cost, CO2).
Main methods	create, status, download

SchedulerService — Timed jobs (BullMQ repeatable)

Responsibilities	• Every 30 s: device and meter offline check. Every minute: window-rule evaluation, timed-suspension lift, escalations. Hourly: invite/OTP clean-up. Nightly: shift instance generation, export clean-up, usage counters.
Main methods	register jobs

AuditService — Traceability

Responsibilities	• Interceptor records every create/update/delete with before/after (secrets removed); queryable by Tenant Admin and Auditor.
Main methods	record, list

DeviceCommandService — Commands to a device

Responsibilities	• Send ping, get info, sync time, set polling, read Modbus registers (read only), clear stored readings, restart. • Queued → Sent → Success / Failed / Timed out (10 s); replies stored with the command; restart / clear need confirmation and are audited.
Main methods	send, list, get, onReply, timeout

PublishService (worker) — Forwarding readings to other systems

Responsibilities	• CRUD publish destinations per device (MQTT broker or HTTPS webhook, topic/URL, interval, payload format, fields, meters, on/off). • Builds each payload from the live cache at its interval, retries twice, records every attempt; Test publish; payload preview; 'Publish failing' alert after 3 failures.
Main methods	createDestination, updateDestination, removeDestination, test, preview, runInterval

TelemetryLogService — Device telemetry log

Responsibilities	• Writes every message per meter and polling cycle with status (ok, late, timeout, crc, out of range), latency and raw payload; kept 7 days. • Serves the device's Telemetry tab (log + chart) and the Data logs tab (publish attempts and command replies).
Main methods	record, list, chartSeries, dataLogs

TariffInstanceService — Runs of Time-of-Day slots

Responsibilities	• Creates the next 2 days of tariff instances from the slabs (Mon–Sat T1–T5, Sunday Flat; T1 22:00 → 06:00 counted on its start day). • Adds live energy to the running instance, closes it from 15-minute aggregates; totals and By-slot summary for Energy cost → Instances.
Main methods	generate, onReading, close, list, summary, get

4. APIs

Base URL `/api/v1`, JSON, Authorization: Bearer <access token> (refresh token in an httpOnly cookie). The tenant always comes from the token. Paging: `?page=&size=` → `{ data, page, size, total }`. Errors: `{ error: { code, message, field? } }` — VALIDATION, UNAUTHENTICATED, FORBIDDEN, NOT_FOUND, CONFLICT, LIMIT_REACHED, RATE_LIMITED. Realtime updates use the Socket.IO namespace `/live` (events: reading, meter.status, device.status, dashboard, alert, notification, session.revoked).

4.1 Auth

Method	Path	Purpose
POST	<code>/auth/register</code>	Register a tenant + admin (→ pending)
POST	<code>/auth/verify-email</code>	Confirm the 6-digit code
POST	<code>/auth/login</code>	{orgId, login, password, remember} → token, or {challenge:'otp'}
POST	<code>/auth/otp/request</code> · <code>/auth/otp/verify</code>	OTP sign-in / 2FA step
POST	<code>/auth/refresh</code>	Rotate tokens (cookie)
POST	<code>/auth/logout</code>	Revoke this session
POST	<code>/auth/forgot-password</code> · <code>/auth/reset-password</code>	Reset flow
GET	<code>/auth/me</code>	User, tenant, role permissions, field permissions, limits
GET	<code>/auth/memberships</code>	Other tenants with the same email
POST	<code>/auth/switch-tenant</code>	{orgId}
GET · DELETE	<code>/auth/sessions/:id</code>	Active sessions / sign out a device
GET · POST	<code>/invites/:token[/accept]</code>	Show and accept an invitation

4.2 Control Center (platform realm)

Method	Path	Purpose
POST	<code>/admin/auth/login</code>	Platform admin sign-in (+ MFA)
GET	<code>/admin/tenants?status=</code>	Tenants and approvals
GET	<code>/admin/tenants/:id</code>	Profile, usage, limits, history
POST	<code>/admin/tenants/:id/approve</code> · <code>/reject</code> · <code>/suspend</code> · <code>/reactivate</code>	{reason, until}
PUT	<code>/admin/tenants/:id/limits</code>	{users: 60, devices: 0, reason}
GET	<code>/admin/tenants/:id/users</code>	Read-only list
DELETE	<code>/admin/tenants/:id</code>	Soft delete

4.3 Tenant, users, roles, sites

Method	Path	Purpose
GET · PATCH	<code>/tenant</code>	Organisation profile
GET	<code>/tenant/limits</code>	Limit, used and remaining per key
GET · POST	<code>/users</code>	List (filters, paging) / invite
GET · PATCH · DELETE	<code>/users/:id</code>	
POST	<code>/users/:id/resend-invite</code> · <code>/suspend</code> · <code>/activate</code>	
GET · POST	<code>/roles</code>	
PATCH · DELETE	<code>/roles/:id</code>	
GET · PUT	<code>/roles/:id/permissions</code>	Module × action matrix
GET · PUT	<code>/roles/:id/field-permissions</code>	module.field → {view, edit}

Method	Path	Purpose
GET · POST · PATCH · DELETE	/sites[:id]	
GET · POST	/permissions	Permission key catalogue
GET · POST · PATCH · DELETE	/cost-centres[:id]	
PATCH · POST	/me · /me/password	My profile / Change password

4.4 Devices, meters, connections

Method	Path	Purpose
GET · POST	/devices	POST returns { device, token } — the token only this once
GET · PATCH · DELETE	/devices/:id	
GET	/devices/:id/meters	Meters it reads, with slave IDs
POST	/devices/:id/token/regenerate	New device token (shown once); old token revoked
GET	/devices/:id/telemetry?from&to&slave&status&page&size	Telemetry log (+ CSV with Accept: text/csv)
GET · POST	/devices/:id/commands	Command history / send a command
GET	/devices/:id/commands/:cmdId	Status and reply
GET · POST	/devices/:id/publish-destinations	
PATCH · DELETE	/devices/:id/publish-destinations/:destId	Edit, enable / pause, delete
POST	/devices/:id/publish-destinations/:destId/test	Test publish now
GET	/devices/:id/publish-destinations/:destId/preview	Next payload
GET	/devices/:id/data-logs?from&to&kind&status&q&page&size	Publishes + command replies
GET	/devices/:id/events	Status history
GET	/meters?view=tree&site=&status=&kind=	Meter tree (replaces ED_METERS)
POST	/meters	Create main meter
POST	/meters/:id/sub-meters	Create sub-meter
GET · PATCH · DELETE	/meters/:id	Delete = archive
GET	/meters/:id/live	Latest values from the live cache
GET	/meters/:id/readings?from&to&interval=auto 1m 15m 1h 1d¶ms=kw,pf	Chart series
GET	/meters/:id/summary?period=today month	kWh, cost, peak, PF, Σ/Δ
GET · PUT	/connections/flow	Current draft
POST	/connections/flow/validate · /deploy	
GET	/connections/flow/versions	
POST	/connections/flow/versions/:v/rollback	

4.5 Analytics, tariffs, shifts

Method	Path	Purpose
GET	/dashboard/summary	Tiles on the Dashboard
GET	/analytics/consumption?from&to&groupBy=meter site shift day	
GET	/analytics/sub-meter-report?from&to	Sub-meter consumption report
GET	/analytics/demand?month=	15-min maximum demand vs contract
GET	/analytics/power-quality?from&to	PF, THD, voltage, imbalance
GET · PUT	/energy-targets/:meterId/:month	Target & allocation

Method	Path	Purpose
GET · POST · PATCH · DELETE	/tariffs/{:id}	ToD slots
GET	/tariffs/active?at=	Slot and rate now
GET	/tariff-instances?from&to&slot&status&page&size	Tariff instances list
GET	/tariff-instances/summary?from&to	Totals and By-slot table
GET	/tariff-instances/{:id}	Detail with per-meter split
GET · POST · PATCH · DELETE	/shifts/{:id} · /shift-schedules/{:id}	
GET · POST · DELETE	/shift-assignments	Assign staff
GET	/shift-instances?from&to	
GET	/shift-instances/{:id}/energy	Energy, cost, peak per meter

4.6 Alerts, notifications, exports, audit

Method	Path	Purpose
GET	/alerts?status&severity&category&search&page&size	Alert log
POST	/alerts/{:id}/acknowledge · /resolve	{note}
POST	/alerts/acknowledge-all	
DELETE	/alerts/{:id}	Tenant Admin only
GET · POST	/alert-rules?meter=	
GET · PUT · DELETE	/alert-rules/{:id}	
POST	/alert-rules/{:id}/test	24-hour dry run
GET	/notifications?unread=true	Bell inbox
POST	/notifications/{:id}/read · /read-all	
GET · PUT	/me/notification-preferences	
POST	/exports	{kind, format, from, to, meters}
GET	/exports/{:id} · /exports/{:id}/download	
GET	/audit?entity&actor&from&to	Tenant Admin, Auditor

4.7 Device ingestion (device credentials, not user tokens)

Method	Path	Purpose
POST	/ingest/v1/readings	Batch of readings (HTTPS push gateways)
POST	/ingest/v1/heartbeat	Status, signal, firmware
POST	/internal/mqtt/auth · /internal/mqtt/acl	Called by EMQX to authorise devices (internal network only)